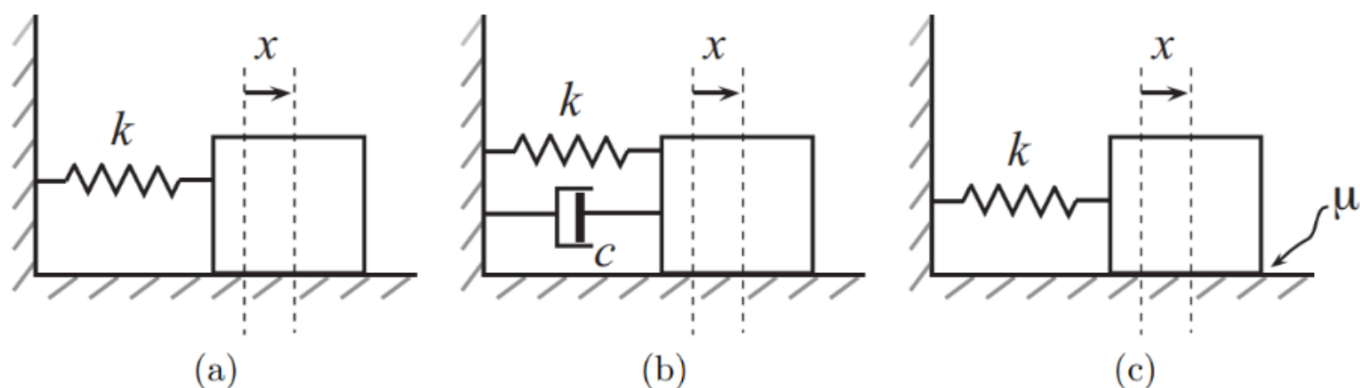




Plot the 2D phase portrait for the three spring-mass systems shown, identifying all important features; the generalized coordinate, x , is the stretch in the spring.

(a) Sliding without friction along the floor.

(b) With a linear damper $F = -b\dot{x}$.

(c) With Coulomb (kinetic) friction due to contact between mass and floor. Coulomb friction can be found by $F = -\mu N \frac{\dot{x}}{|\dot{x}|}$.

a)

$$L = \frac{1}{2} M \dot{x}^2 - \frac{1}{2} k (x^2 - x_0^2)$$

$$M \ddot{x} + kx = 0$$

$$v = \dot{x}$$

$$\dot{x} = v$$

$$0, 0$$

$$\dot{v} = -\frac{k}{m} x$$

$$\dot{v} = -\frac{k}{m} x$$

$$J = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{bmatrix}$$

$$\tau = 0$$

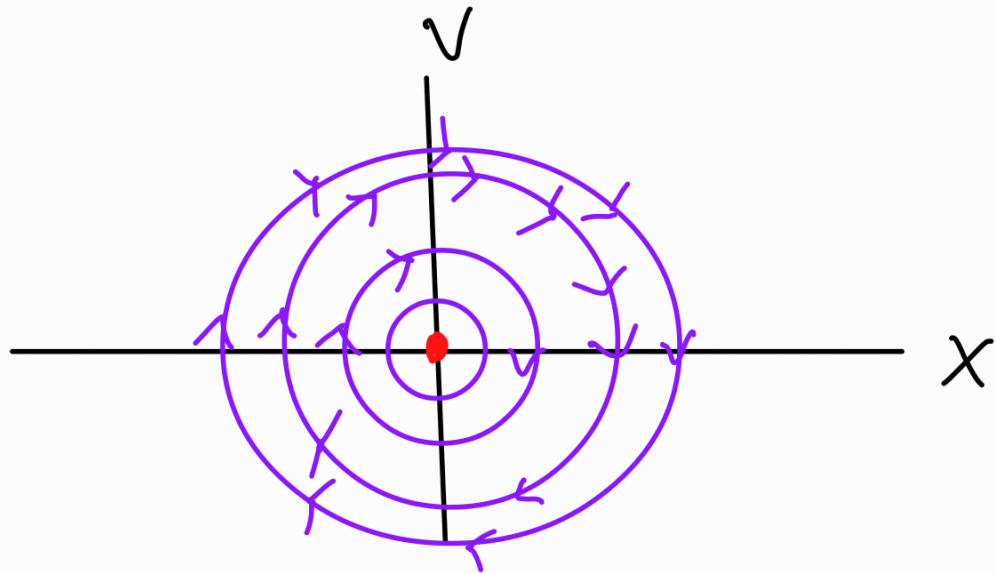
$$\Delta = k/m$$

$$\lambda = \frac{\text{tr}(J) \pm \sqrt{\text{tr}(J)^2 - 4 \det(J)}}{2}$$

$$\lambda = \frac{\pm \sqrt{-4de + (5)}}{2}$$

$$\lambda = \pm \sqrt{k/m} i$$

Pure imaginary = center



b)

$$\sum f = \sum ma$$

$$-kx - b\dot{x} = m\ddot{x}$$

$$\ddot{x} = -\frac{b}{m}\dot{x} - \frac{k}{m}x$$

$$\dot{x} = v$$

(0,0)

$$\dot{v} = -\frac{b}{m}v - \frac{k}{m}x$$

$$J = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix}$$

$$\tau = -b/m$$

$$\Delta = k/m$$

$\tau < 0 \therefore$ stable

$$\frac{b^2}{m^2} - \frac{4k}{m} = 0$$

$$b = 2\sqrt{km}$$

$b > 2\sqrt{km} =$ stable node

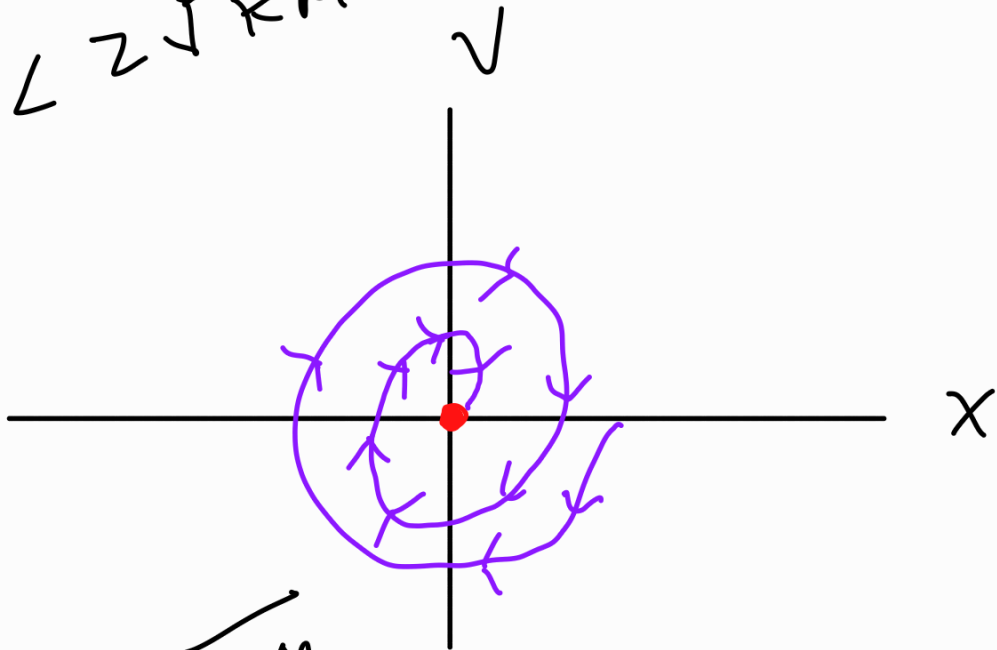
$b < 2\sqrt{km} =$ stable spiral

$$\begin{bmatrix} -\lambda & 1 \\ -\frac{k}{m} & -\frac{b}{m} - \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

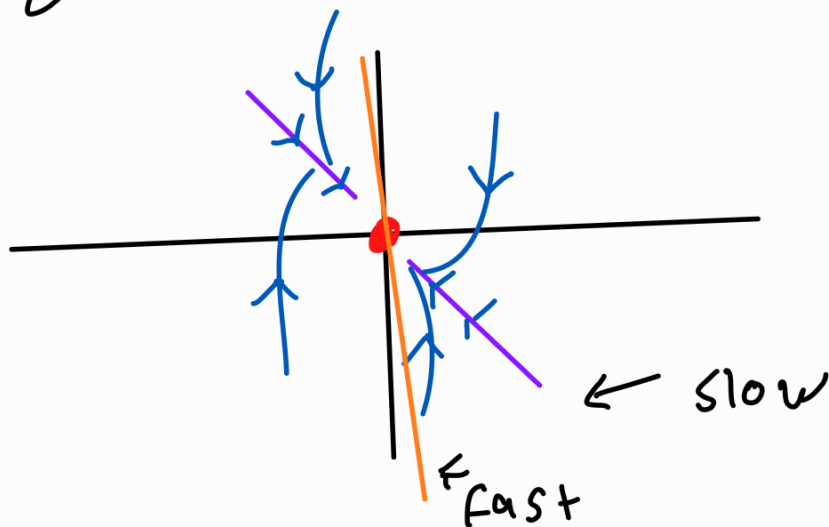
$$v_1 \lambda = v_2 \therefore v = \lambda x$$

λ is negative

$$b < 2\sqrt{km}$$



$$b > 2\sqrt{km}$$



$$\dot{x} > 0$$

C)

$$\ddot{x}_M = -\kappa x - \mu N$$

$$\ddot{x}_M = -\kappa \left(x + \frac{\mu}{\kappa} N \right)$$

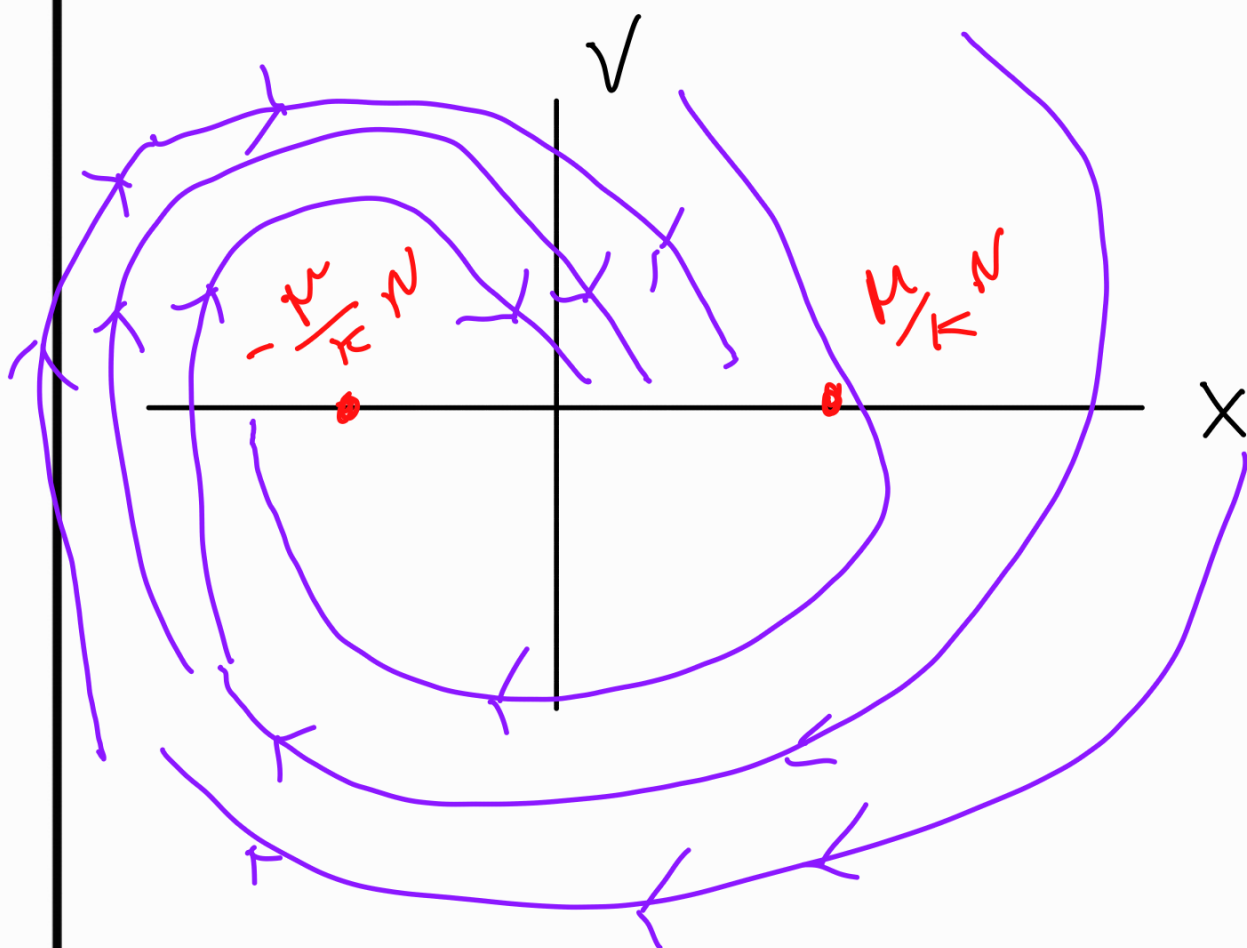
$$\text{zero: } -\frac{\mu}{\kappa} N$$

$$\dot{x} < 0$$

$$\ddot{x}_M = -\kappa x + \mu N$$

$$= -\kappa \left(x - \frac{\mu}{\kappa} N \right)$$

$$\text{zero: } \frac{\mu}{\kappa} N$$



Problem 2

Reconsider the rotating bead of mass m from Homework 1. It is constrained to slide without friction in the tube which is welded to a vertical post at the fixed angle θ . The post rotates with constant angular velocity Ω .

(a) Find a differential equation for the variable r when $\theta = 60^\circ$, using Lagrange's equations this time.

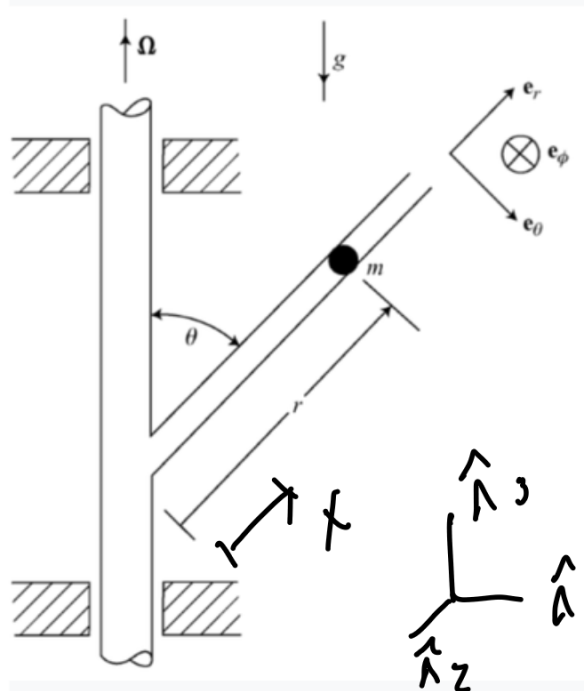
(b) Non-dimensionalize the equation of motion.

(c) Plot the phase portrait of the non-dimensionalized system, identifying all important features. Only show the physically relevant portion of the phase space.

(d) If the bead starts at $r_0 = \frac{4g}{3\Omega^2}$, what minimum initial inward speed $\dot{r}_0$ does it need in order to eventually reach the bottom of the tube, $r = 0$?

Note: You don't need to directly solve for $r(t)$ to do this. Instead, consider the Hamiltonian (the more general definition, not the total energy), which is a constant of motion, and find the initial conditions that have the same value of H as the equilibrium. This will find the location of the stable manifold at

$$r = \frac{4g}{3\Omega^2}.$$



gen coords : x

$$\frac{t_2}{t_1} \propto \frac{1}{\theta}$$

$$r \propto \theta^3$$

$$\vec{r} = x \hat{e}_r$$

$$\dot{\theta} \cos \theta \hat{e}_r - \dot{\theta} \sin \theta \hat{e}_\theta$$

$$\dot{\vec{r}} = \dot{x} \hat{e}_r + \dot{\theta} \hat{e}_3 \times x \hat{e}_r$$

$${}^N R_E = \begin{bmatrix} \sin \theta & \cos \theta \\ \cos \theta & -\sin \theta \end{bmatrix} = {}^N R_E$$

$$\dot{\vec{r}} = \dot{x} \hat{e}_r + \dot{\theta} x \sin \theta \hat{e}_3$$

$$|\dot{\vec{r}}|^2 = \dot{x}^2 + \dot{\theta}^2 x^2 \sin^2 \theta$$

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{\theta}^2 x^2 \sin^2 \theta)$$

$$V = mgx \cos \theta$$

$$L = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} \dot{\theta}^2 x^2 \sin^2 \theta - mgx \cos \theta$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{a}} \right) - \frac{\partial L}{\partial a} = 0$$

a) $m \ddot{x} - \Omega^2 m \sin^2 \theta_0 x + mg \cos \theta_0 = 0$

b) $\ddot{x} = \Omega^2 \sin^2 \theta_0 x - g \cos \theta_0$

$$x = L \psi \quad t = T \phi \rightarrow \frac{dx}{dt} = \frac{L}{T} \frac{d\psi}{d\phi}$$

$$\frac{d^2 x}{dt^2} = \frac{L}{T^2} \frac{d^2 \psi}{d\phi^2}$$

$$\frac{L}{T^2} \ddot{\psi} = \Omega^2 \sin^2 \theta_0 L \psi - g \cos \theta_0$$

$$\tau_a^0$$

$$\ddot{V} = T^2 \Omega^2 \sin^2 \theta_0 V - \frac{T^2 g}{L} \cos \theta$$

$$T^2 \Omega^2 \sin^2 \theta_0 = 1$$

$$T = \sqrt{\frac{1}{\Omega^2 \sin^2 \theta_0}}$$

$$\frac{\frac{1}{\Omega^2 \sin^2 \theta_0} g \cos \theta}{L} = 1$$

$$L = \frac{g \cos \theta}{\Omega^2 \sin^2 \theta}$$

$$\ddot{V} = V - 1 \quad L|_{\theta=60} = \frac{1}{2} g / \Omega^2 \frac{3}{4} = \frac{1}{2} g \cdot \frac{4}{3 \Omega^2} = \frac{4}{6} \frac{g}{\Omega^2} = \frac{2}{3} \frac{g}{\Omega^2}$$

$$\ddot{v} = v - 1 \quad \dot{v} = v \quad \text{zeros:}$$

$$\dot{v} = v - 1 \quad (1, 0)$$

$$J = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \tau = 0$$

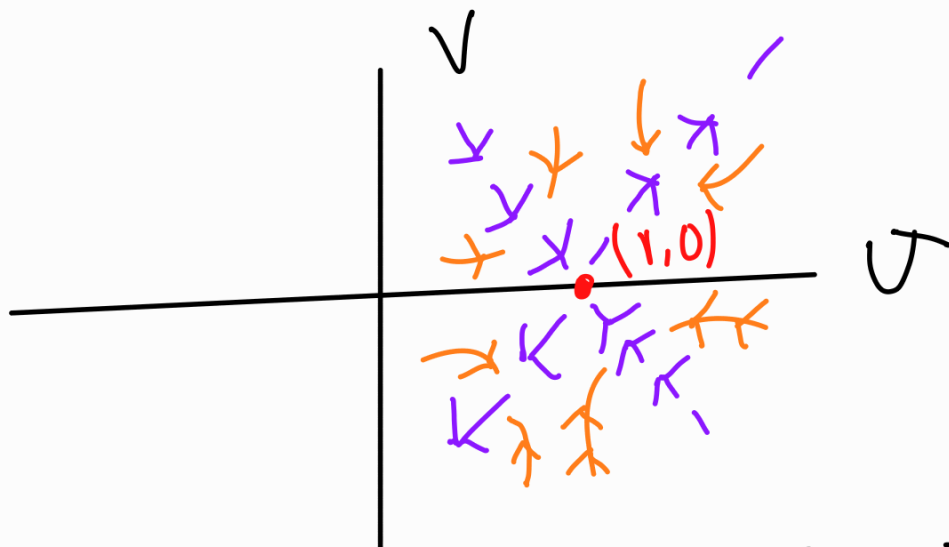
$$\Delta = -1$$

$\therefore$ saddle point

$$\lambda = \pm 1 \quad v = v - 1 \quad \leftarrow \text{stable}$$

$$v = -(v - 1)$$

c)



$$\int (\ddot{v} = v - 1) \quad \dot{v} = \frac{1}{2} v^2 - \frac{1}{2} (v - 1)^2$$

$$= C$$

$$r_0 = \frac{4g}{3\Omega^2} = 2 \text{ m}$$

$$\frac{4g}{3\Omega^2} = \frac{2}{3} \frac{g}{\Omega^2} \text{ m} \therefore \Omega = 2$$

$$V = -(\psi - 1) \big|_{\psi=2} \therefore V_0 = -1$$

$$\frac{L}{T} = \frac{\frac{g \cos \theta}{\Omega^2 \sin^2 \theta}}{\frac{1}{\Omega \sin \theta}} = \frac{g}{\Omega \tan \theta}$$

d)

$$\dot{x}_0 = \frac{L}{T} V_0 = \frac{-g}{\Omega \tan \theta}$$

$$H = \frac{1}{2} v^2 - \frac{1}{2} (v-1)^2 \Big|_{v_0=2} \quad \text{at } t=0$$

$$H = \frac{1}{2} v_0^2 - \frac{1}{2}$$

↓

$$v^2 = 2H + (v-1)^2$$

$$v^2 \Big|_{v=1} = 2H \geq 0$$

$$v^2 \Big|_{v=1} = v_0^2 - 1 \geq 0$$

$$|v_0| \geq 1$$

$$v_0 \leq -1$$

$$\dot{x} = \frac{L}{T} v_0 = -\frac{g}{L \tanh \theta}$$

Problem 3

Consider a mass-spring system, with $m = 1$ and $k = 1$, that is experiencing a damping force of $F = \mu(1 - x^2)\dot{x}$.

(a) Write its equation of motion, $\ddot{x} = f(x, \dot{x}; \mu)$.

(b) Convert to state-space by using $v = \dot{x}$ to write the two first-order ODEs $\dot{x} = f_1(x, v; \mu) = v$, $\dot{v} = f_2(x, v; \mu)$.

(c) Choose $\mu = -1$ and draw the phase portrait. Where is the one equilibrium in the system?

(d) Linearize the system about $(x, v) = (0, 0)$ using the Jacobian,

$$\begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial v} \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}$$

(e) Find the eigenvalues of the Jacobian. Is the equilibrium point stable for $\mu = -1$?

(f) Now choose $\mu = 1$ and find the eigenvalues of the Jacobian matrix again. Is the equilibrium point stable now?

(g) Draw the phase portrait again. What has changed about the system?

a) $m\ddot{x} = -kx + \mu(1 - x^2)\dot{x}$

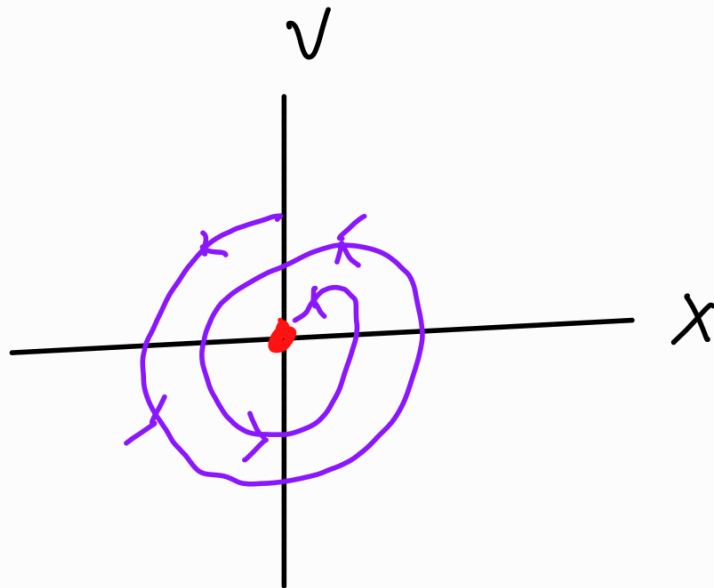
$$\ddot{x} = -\frac{k}{m}x + \frac{\mu}{m}(1 - x^2)\dot{x}$$

b) $\dot{x} = v$

$$\dot{v} = -\frac{k}{m}x + \frac{\mu}{m}(1 - x^2)v$$

c) $\mu = -1 \therefore \begin{cases} \dot{x} = v \\ \dot{v} = -x - v(1 - x^2) \end{cases}$

fixed point: $(0, 0)$



d)

$$\begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 + 2vx & -1 + x^2 \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}$$

$$= \begin{matrix} 0,0 \\ \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} \end{matrix}$$

$$\tau = -1 \quad \det E = 1$$

$$e) \quad \lambda = \frac{+r \pm \sqrt{+r^2 - 4\Delta}}{2}$$

$$-\frac{1}{2} \pm \frac{\sqrt{1-4}}{2} \quad \checkmark -3$$

$$-\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$$

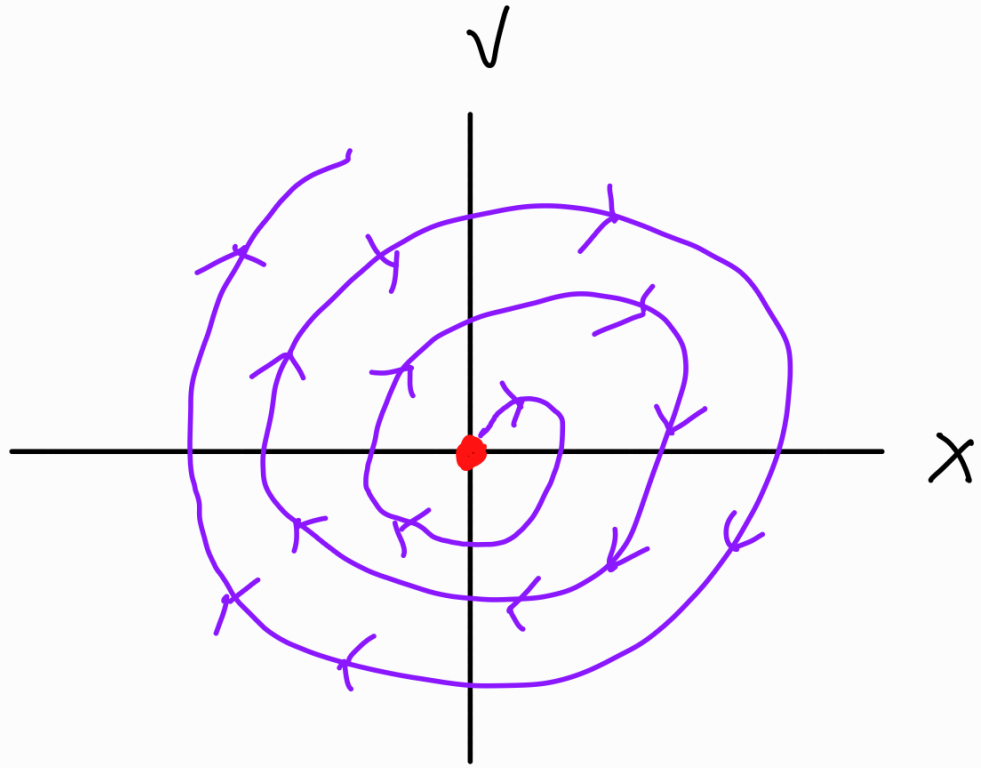
$\text{Re}(\lambda)$ both negative
stable spiral

$$f) \quad J = \begin{bmatrix} 0 & 1 \\ -1 & \mu \end{bmatrix} \Big|_{\mu=1} \quad \begin{matrix} \text{Tr} = 1 \\ \Delta = 1 \end{matrix}$$

$$\lambda = \frac{1}{2} \pm \frac{\sqrt{3}}{2} i$$

$\text{Re}(\lambda) > 0$, unstable spiral

g)



Problem 4

Consider a mass hanging vertically from a nonlinear spring, with a total potential energy given by

$$V(y) = -mgy - \frac{1}{2}k_1y^2 + \frac{1}{4}k_2y^4$$

When k_1 is positive, the spring is repelling for small values of y , and when it is negative, it is attracting. y is defined positive downward. This is similar to the pitchfork bifurcation example that we saw in 2D.

The system also has a damping force, $F_d = -b\dot{y}$. This gives the equation of motion,

$$m\ddot{y} = -b\dot{y} - \frac{dV}{dy}$$

(a) Nondimensionalize the system. Show that by choosing your time and length scalings to be $T = \frac{m}{b}$ and $L = \frac{b}{\sqrt{k_2}}$, you reduce to the system,

$$\ddot{x} + \dot{x} - ax + x^3 = h,$$

where $x = \frac{y}{L}$. What are a, h ?

(b) Plot two phase portraits, one with $h=0.01$, $a=-1$, and the other with $h=0.01$, $a=1$. What do you notice about those phase portraits?

(c) Convert the system to (x, v) state space. Equilibria occur only when $v^* = 0$. What is the cubic polynomial that needs to be solved to find x^* ?

(d) Analytically calculate the Jacobian of the system.

(e) Using a polynomial root solver (like roots in MATLAB or numpy.roots in Python), generate a scatter plot of all real roots of the system over the range $a \in [-1, 1]$ for $h=0.01$. Use an eigenvalue solver to solve for the eigenvalues of the system to classify stability and color the roots according to their stability. You'll notice that this is a perturbed version of the pitchfork bifurcation. Because the addition of a small asymmetry (such as $h=0.01$) breaks the pitchfork bifurcation, the bifurcation itself is not stable.

$$\begin{aligned} y &= Lx & t &= Ts & \frac{dy}{dt} &= \frac{L}{T} \frac{dx}{ds} \\ \frac{dV}{dy} &= -mg - k_1y + k_2y^3 & \frac{d^2y}{dt^2} &= \frac{L}{T^2} \frac{d^2x}{ds^2} \\ \frac{mL}{T^2} \ddot{x} &= -\frac{bL}{T} \dot{x} + mg + k_1Lx - k_2L^3x^3 \\ \ddot{x} &= -\frac{b}{m} T \dot{x} + \frac{gT^2}{L} + \frac{k_1T^2x}{m} - \frac{k_2L^2T^2x^3}{m} \\ \ddot{x} + \frac{b}{m} T \dot{x} - \frac{gT^2}{L} - \frac{k_1T^2x}{m} + \frac{k_2L^2T^2x^3}{m} &= 0 \end{aligned}$$

$$\frac{b}{m} T = 1 \quad \therefore T = \frac{m}{b}$$

$$\frac{k_2}{m} L^2 T^2 = 1 \quad \Big|_{T = m/b}$$

$$L^2 k_2 \frac{m}{b^2} = 1$$

$$L = \sqrt{\frac{b^2}{k_2 m}} = \frac{b}{\sqrt{k_2 m}}$$

$$\ddot{X} + \dot{X} - \frac{k_1 m}{b^2} X + X^3 = \frac{g \sqrt{k_2} m^{5/2}}{b^3}$$

$$a = \frac{k_1 m}{b^2} \quad h = \frac{g \sqrt{k_2} m^{5/2}}{b^3}$$

$$\frac{g T^2}{L} = \frac{g \frac{m^2}{b^2}}{\frac{b}{\sqrt{k_1 m}}}$$

$$\frac{g m^2}{b^2} \cdot \frac{\sqrt{k_1 m}}{b} = \frac{g m^{5/2} \sqrt{k_2}}{b^3}$$

b) see latex

c)

$$\dot{x} = v$$
$$\dot{v} = -v + \frac{k_1 m}{b^2} x - x^3 + \frac{g \sqrt{k_2} m^{5/2}}{b^3}$$

for fixed point, $v = 0$ (first eq constraint)
 $\therefore$

$$x^3 - \frac{k_1 m}{b^2} x - \frac{g \sqrt{k_2}}{b^3} m^{5/2} = 0$$

d)

$$J = \begin{bmatrix} 0 & 1 \\ \frac{k_1 m}{b^2} - 3x^2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ a - 3x^2 & -1 \end{bmatrix}$$

e) see latex